Preventing Particle Penetration

With residents' health increasingly dependent on a home's indoor air quality, ventilation strategies become more important than ever.

by Dara Bowser and Don Fugler

As anyone who has stood over a smoky campfire can attest, breathing in fine particles is a health hazard. The health risk associated with inhaling dust and other small particles is directly related to the level of exposure, which is why agencies such as the Ontario Ministry of the Environment have set up outdoor atmospheric sampling stations to measure the levels of fine particles in outdoor air. But given that people spend most of their time indoors, indoor particle exposure may be more important than outdoor exposure (see "Sources of Indoor Particles," p. 14).

Unfortunately, there is little or no information now about how different ways of building, heating, cooling, and ventilating a home affect how well it guards the occupants from fine particles coming in from the air outside. With asthma, allergies, and other respiratory ailments that are linked with indoor air quality (IAQ) on the rise, better IAQ means better health for consumers. And knowing what ventilation strategies provide the best IAQ can help consultants, contractors, and builders to design and build healthier homes. Canada Mortgage and Housing Corporation (CMHC) is the federal agency responsible for housing research. CMHC hired Dara Bowser of Bowser Technical, Incorporated, to provide some of that information by investigating how ventilation and filtration affect the indoor/outdoor ratio of particles in a house, and by determining the filtering effect of the house envelope on incoming air.


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Outdoor Particle Penetration

Outdoor particles can penetrate either through the building envelope or through an intentionally created air inlet. The paths range from the relatively large and direct, such as—in Canada at least—the combustion air inlet for gas furnaces and water heaters, which are usually 4–6-inch open pipes, to the lengthy and convoluted, such as paths through the wall assembly. The pressure causing the air movement may be due to naturally occurring forces, such as wind and indoor/outdoor temperature differences; or the pressure may be caused by the operation of mechanical equipment within the home, such as the outdoor air supply connection to an air handler, an exhaust-only ventilation system, or a clothes dryer.

In Canada, some ventilation systems consist of an intake duct connected to the return air side of the furnace. These are sometimes called nonpowered systems, but in reality the operation and suction pressure created by the furnace fan governs the rate of intake air flow. Other ventilation systems are based on a central heat recovery ventilator (HRV). HRVs are equipped with filters that are capable of removing large particles that would block the heat exchange passages, but these filters are not usually effective for removing fine particles from the airstream.

When a central forced-air system is also installed in a home, the HRV often has its outdoor air supply connected to the return air duct of the central forced-air system. Internal fans and controls manage the HRV flow rate. In both of these systems, the incoming outdoor air is passed through the central forced-air system's main filter before being delivered into the occupied spaces of the home. The various filters in central forced-air systems remove anywhere from none of the fine particles to more than 90% of them.

Testing Ventilation Strategies

Our experiments took place in a home in Brantford, Ontario, Canada. Two adults lived in the house at the time of our tests. The test conditions represent typical southern Ontario spring and summer conditions. During our tests, the air handling and ventilation systems were operated continuously, with all the doors and windows closed. The continuous operation of ventilation systems is becoming more popular in Canada as the housing stock becomes more airtight, residents become more aware of IAQ, and summer air conditioning becomes more prevalent. Also, public health officials recommend that people who have respiratory problems caused by outdoor air particles keep their doors and windows closed. The Brantford house is considered to be moderately airtight by Canadian standards, with an effective leakage area (ELA) of 114 square inches (733 cm^2) and an ACH_{50} of 5.33.

The Brantford house consists of a basement and an upper floor. The upper floor contains the bedrooms, living room, and kitchen. The basement contains the mechanical room and a home office. We located the instruments for the experiment in the mechanical room. One person was in the home during the day (mostly in the basement). Two people were in the home during the evening and at night (mostly on the upper floor).

We measured airborne fine particles

- outside (under the overhanging roof of a covered patio);
- in the air intake from outside;
- in the home office;
- in the central air handling system (the sample consisted of mixed return and outside air); and
- in one bedroom.

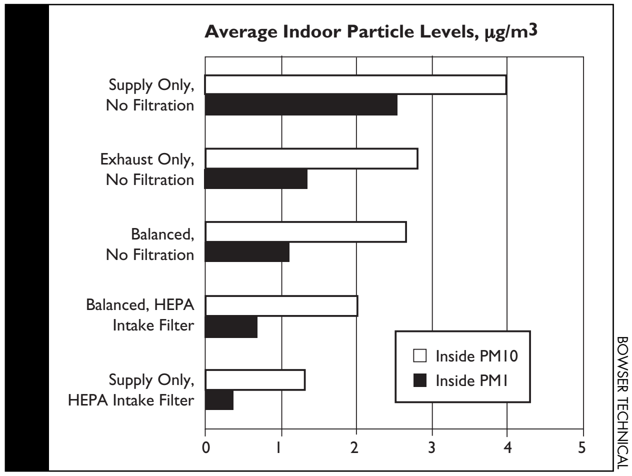


Figure 1. Indoor particle levels were substantially higher for the Supply Only, No Filtration mode than for any of the other modes. This was true for both PM1 and PM10 size ranges.

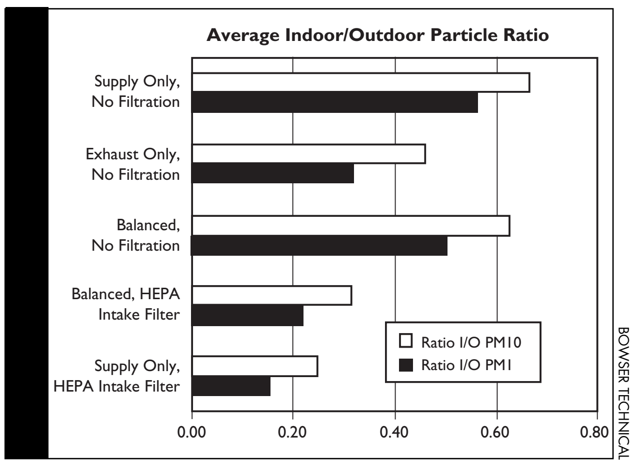


Figure 2. When the data are expressed in terms of the indoor/outdoor particle ratio, it can be seen that there are distinct differences between the Balanced, No Filtration mode, and the Exhaust Only, No Filtration mode that are not evident when observing indoor particle level alone. These differences are caused by the high level of variability in outdoor particle level from experiment to experiment.

We tested for fine-particle penetration for five distinct ventilation modes. These were

- Supply Only, No Filtration;
- Exhaust Only, No Filtration;
- Balanced, No Filtration;
- Balanced, with HEPA Intake Filter; and
- Supply Only, with HEPA Intake Filter.

Ventilation rates ranged between 1.20 and 0.71 ACH and were selected to ensure that in the Supply Only modes, all of the incoming air passed though the ventilation system, and in the Exhaust Only mode, all of the incoming air passed through the building envelope. Continuous real-time measurements of indoor and outdoor particle levels were made during the day and night. We also measured air temperature, air pressure, wind speed, and ventilation flows continuously. A total of 428 data hours were used for data analysis.

We calculated a house average value using the average of the office and bedroom values. Indoor particle levels were reported as the amount of particles less than 1 micron (PM1) and as the amount of particles less than 10 microns (PM10). Indoor/outdoor ratios were also reported. An indoor/outdoor ratio of 0.5 indicates, for example, that the indoor levels were 50% of the outdoor levels. A ratio of 1.5 indicates that the indoor levels were 150% of the outdoor levels.

Comparing Ventilation Strategies

In general, the systems with HEPA filtration of the outdoor air provided the best control of indoor particles coming from outdoors. The systems that supplied outdoor air directly to the living space without filtration didn't do well in keeping particles out. The Exhaust Only system, which relies on the building envelope to filter out incoming particles, performed in the middle, halfway between the systems with HEPA filtration and the other two systems without filtration (see Figures 1 and 2).

The dynamic functioning of the systems can be seen in sample data sets. With the Supply Only HEPA and the Balanced HEPA systems respectively, the indoor particle levels compared to the outdoor levels show only a mild trend. The indoor particle levels appear to be related principally to occupant activity, rather than to outdoor levels.

With the Exhaust Only system, there is a distinct relationship between indoor particle levels and outdoor particle levels, but there is a significant time lag between the indoor and outdoor levels, and the effects of occupant activity remain apparent (see Figure 3). With the Supply Only, No Filtration system, indoor particle levels are essentially a mirror of outdoor particle levels, with a lag time that is attributable to the settlement rate of the indoor particles (see Figure 4). The effect of outdoor particle penetration is much greater than the effect of occupant activity.

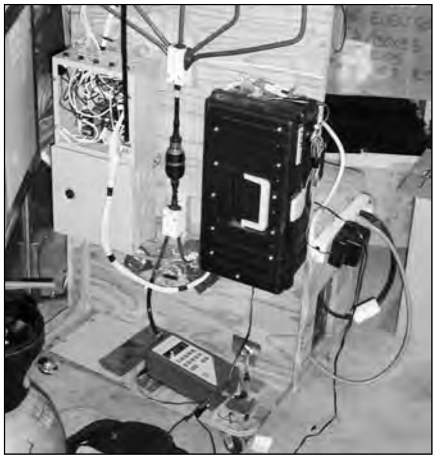
Recommended Ventilation Strategies

From our measurements of particle penetration into a house under various ventilation conditions, we can offer some practical advice.

Direct entry of unfiltered air. The direct entry of air that is not filtered for particles will raise the indoor levels of respirable particles. These levels will be higher than those normally expected in a home that does not have direct entry of outdoor air, or in which the entry routes for outdoor air are filtered. This is true even in comparison to Exhaust Only ventilation systems with similar air change rates, as the building envelope of the house is capable of removing a substantial portion of the incoming fine particles, even those with relatively small diameters.

For houses that are currently equipped with an HRV-based system that is not connected to a central forced-air recirculation system, a fine-particle filtering system should be considered for the incoming air. For homes equipped with an outdoor air intake directly connected to the return of a forced-air system, or where an HRV is connected to the forced-air system, the use of a fine-particle air filter for the central system should be considered. Such a fine-particle filter will treat incoming outdoor air as well as recirculated house air.

Exhaust Only ventilation. Exhaust Only ventilation systems are moderately effective at keeping fine particles out of the living spaces of a home, due to the



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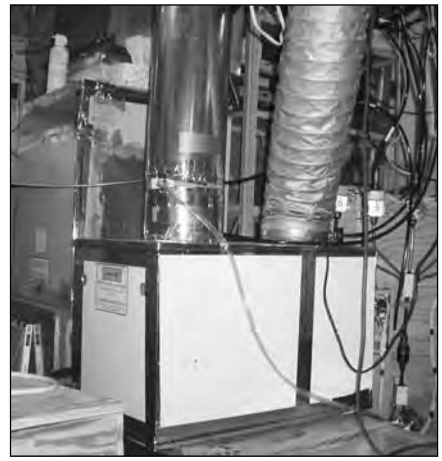

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(left) This view of the particle counting rig shows 5 continuous sampling intakes converging into a single sampling stream. The air samples pass first through an impact separator and then into the optical particle counter. The timing mechanisms and solenoid valves "time-share" the sampling stream with each of the 5 sampling intakes. *(middle)* Outdoor air pressure sampling took place at a distance from the house. *(right)* The HEPA filter is installed on the intake duct.

Sources of Indoor Particles

The level of indoor fine particles at any given moment is influenced by several factors.

Internal generation. Internal generation of fine particles comes primarily from cooking, and from sources of combustion such as burning candles and smoking. Particles are also generated from human and pet dander. Generation usually occurs when the house is occupied and the occupants are active. Cat dander has been found to have high generation rates and slow settling times.

Resuspension. Resuspension arises from the movement of the occupants in the home during periods of activity. The resuspended particles are stored on surfaces, have low settling rates, and resuspend quickly. They include the internally generated particles mentioned above, particles of outdoor origin that have been tracked in by the occupants, and particles that enter on an infiltration or ventilation airstream.

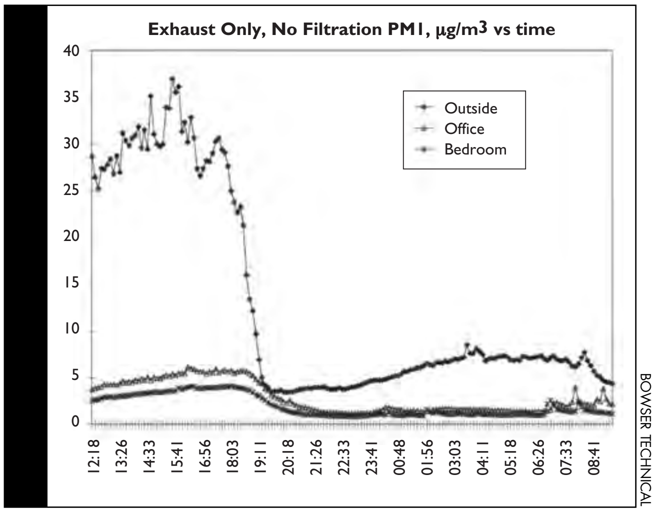
Entry via house envelope. Air entering the living spaces through the house envelope contains a certain number of fine particles. The exact number depends on the level of outdoor fine particles and on how well the building components remove these particles as air passes though them. Some authors (Thatcher and Davis, 1995) have estimated that the filtration effect is negligible, while others (Abt, et al, 2004) have estimated the filtration at anywhere from 6% to 88%, depending on the size of the particle. Outdoor particle levels may vary by a factor of ten or more from day to day and from hour to hour.

filtering effect of the building envelope. While this filtering effect may vary depending on how the house was constructed, it is reasonable to conclude that Exhaust Only systems are superior to those that allow unfiltered air to enter directly into the house. In the absence of combustion equipment that could be depressurized by the action of an Exhaust Only ventilation system, this system could be operated year-round in Canada. The Exhaust Only system can be beneficial during winter operation, because the entry of dry outside air can prevent the wetting of the building envelope by exfiltrating air.

In southern climates, where building interiors are cooler than the outdoor dew point, use of an Exhaust Only system may result in the wetting and subsequent deterioration of the building envelope.

Supply Only ventilation with filtration. Although the Supply Only system with filtration performed better in keeping outdoor particles out of the house, we can't recommended it for wintertime use in heating climates. In the winter, exfiltrating air forced into the building envelope by the operation of such a system could cause wetting of the building envelope, leading once again to deterioration. A Supply Only ventilation system with filtration could be used during the spring, summer, and fall in heating climates and in southern climates, where building interiors are kept cooler and dryer than outdoors.

The pressurization of the building envelope caused by Supply Only ventilation would help to keep the building envelope dry during the summer in a



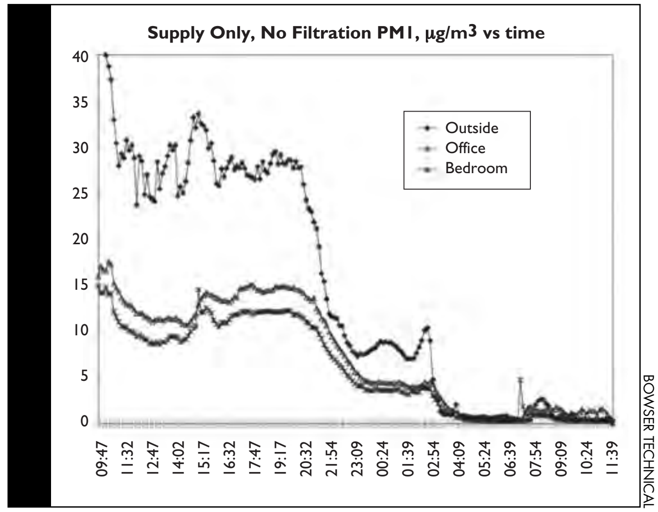


Figure 3. *(left)* With the Exhaust Only system, there is a distinct relationship between indoor particle levels and outdoor particle levels, but there is a significant offset and the effects of occupant activity remain apparent. **Figure 4.** *(right)* With the Supply No filters system, indoor particle levels are essentially a mirror of outdoor particle levels with an offset that is attributable to the settlement rate of indoor particles. The effect of occupant activity is not distinct from the much greater effect of outdoor particle penetration.

Entry via intentional inlet. Air entering the house from outside through an intentionally created inlet will contribute particles to the occupied space depending on the concentration of particles in the outdoor air and the efficiency of the filter placed in the airstream. Efficiencies of currently available filters vary from 0% to 99.5%.

Removal by settling. Particles are removed from the air when they settle on available surfaces in the house. The rate of removal depends on the available surfaces and the settling rate of the individual particles. The operation of an air handling system will increase particle settling by increasing the turbulent surface effect. (It's counterintuitive, but turbulence on the surface of a material can increase the rate of particles settling on it.)

Removal by filtration. A central air handling system equipped with a fine-particle air filter, or a local (in-room) fine-particle air filter may be used to remove particles.

Removal by exhaust and exfiltration. Particles are also removed from a space by air that leaves through the building envelope or through an intentional device such as an exhaust fan. It should be understood that this results in a net reduction of indoor particles only when the air replacing the removed air contains fewer particles than the air it replaces. In many cases the outdoor level of fine particles is higher than the indoor level, so that the exchange of air may actually increase the indoor particle level.

heating climate, or in a southern climate with cool dry inside air.

Balanced ventilation with filtration. The Balanced system with filtration performed almost as well as the Supply Only system with filtration. It would have performed better if the building envelope had been more airtight. A Balanced ventilation system is often chosen for homes in heating climates, where combustion appliances may be affected by the negative pressure induced by an Exhaust Only ventilation system.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A horizontal duct separates the intake (lower right) and the water heater and dryer exhausts, which are located near the corner of the house (left of photo).

Preaching, Perhaps, to the Choir

Many people in the IAQ business assume that the catch phrase "Indoor air is more polluted than outdoor air" is true in all cases. In fact, with respect to fine particles—and especially if you live in an urban area, or where there are other outdoor sources of particulate pollution, such as wood stoves—this is not true. Fortunately, the path to reducing the penetration of outdoor particles is one that we are already following: Increase building airtightness, rely on controlled ventilation, and use sealed-combustion equipment, whenever possible.

Building airtightness can be increased to the point where all, or substantially all, of the ventilation is through a powered duct. In this case, the incoming air can be filtered and the outdoor fine particles removed. We were surprised that the simple addition of a filter to an HRV intake could improve IAQ so dramatically. If you are a professional working in the energy-efficient/healthy house field, this puts one more very important tool in your knowledge toolbox—a tool that you can use to make everyone's life a little better.

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